



Influence of confinement in mesoporous silica on diffusion of a mixture of carbon dioxide and an imidazolium-based ionic liquid by high field diffusion NMR



Eric D. Hazelbaker^a, Rémy Guillet-Nicolas^b, Matthias Thommes^b, Freddy Kleitz^c, Sergey Vasenkov^{a,*}

^a Department of Chemical Engineering, University of Florida, Gainesville, FL 32611, USA

^b Quantachrome Instruments, 1900 Corporate Drive, Boynton Beach, FL 33426, USA

^c Department of Chemistry, Université Laval, 1045 Avenue de la Médecine, Quebec City, QC G1V 0A6, Canada

ARTICLE INFO

Article history:

Received 9 August 2014

Received in revised form 25 November 2014

Accepted 3 December 2014

Available online 10 December 2014

Keywords:

Ionic liquids

Diffusion

NMR

Carbon dioxide

Mesoporous silica

ABSTRACT

Proton and carbon-13 pulsed field gradient (PFG) NMR was applied to study diffusion of the mixture of carbon dioxide and the ionic liquid 1-*n*-butyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide under condition of confinement in the mesopores of KIT-6 silica. The results of these diffusion studies were compared with the results obtained for the corresponding bulk mixture and pure ionic liquid (IL) as well as for the confined pure IL. It was observed that the confinement in mesopores, which were fully loaded with the IL, reduces the cation diffusivity by around a factor of two in comparison to that in the corresponding bulk CO₂/IL mixture or the pure IL. This result was obtained for short diffusion times when possible transport resistances at the external surface of KIT-6 particles are not expected to change the intraparticle diffusivity of the cations. The large experimental uncertainty of the measurements of carbon dioxide diffusivity in the confined CO₂/IL mixture at the smallest diffusion time prevented making a definitive conclusion about the influence of the confinement on CO₂ diffusivity. The reported experimental data suggest that the CO₂ diffusivity under the conditions of confinement can be comparable to or even larger than that in the bulk CO₂/IL mixture. For larger diffusion times, both the carbon dioxide and cation diffusivities in the confined mixture were found to decrease with increasing diffusion time. This diffusion time dependence was attributed to the influence of transport resistances at the external surface of KIT-6 particles.

© 2014 Elsevier Inc. All rights reserved.

1. Introduction

Confinement in porous materials can lead to significant changes of transport properties of ionic liquids (ILs), *i.e.*, molten salts that are liquid at temperatures around room temperature. Studies of such confinement effects become increasingly more important due to development of novel materials such as supported ionic liquid membranes (SILMs) where ILs are in close contact with the pore surface. Owing to the non-volatility and high sorption capacities of ionic liquids for light gases SILMs show promising performance in gas separations [1–6]. In particular, SILM-based separations of CO₂ from mixtures of gases containing CO, N₂ and CH₄ attract much attention. Optimization of such separations requires fundamental understanding of an influence of confinement in pores on dynamics of ILs and mixtures of ILs with gases.

The behavior of ILs in pores of various sizes and types was investigated by molecular dynamics (MD) simulations [7–10]. The simulations revealed that both decrease [7,8] and enhancement [9,10] in the ion mobilities can be caused by the confinement. MD simulations have also demonstrated that gas diffusion in the confined gas/IL mixtures can be faster than that in the corresponding bulk mixtures [9,10]. This observation suggests that for applications in gas capture and separations confined ILs can perform better than bulk ILs. Experimental studies of diffusion of pure ILs in porous materials were performed by microscopic techniques including pulsed field gradient (PFG) NMR [8,11,12] and quasielastic neutron scattering [13]. In a qualitative agreement with the MD simulations discussed above, the results of these studies demonstrate that both increase and decrease in the ion self-diffusivities can be expected as a result of the IL confinement. Macroscopic uptake and permeability measurements of CO₂ showed that the carbon dioxide transport in confined ILs can be faster than that in the corresponding bulk ILs [6,14]. To our knowledge, no studies of

* Corresponding author. Tel.: +1 352 392 0315; fax: +1 352 392 9513.

E-mail address: svasenkov@che.ufl.edu (S. Vasenkov).

gas diffusion in confined ILs were performed by a microscopic technique of any type. Here we report PFG NMR studies of cation and CO₂ self-diffusion in a mixture of CO₂ and an imidazolium-based IL confined in KIT-6 silica [15–17]. KIT-6-type silicas are a large-pore mesoporous materials exhibiting highly ordered three-dimensional (3D) cubic *Ia3d* mesopore structure. KIT-6 was initially synthesized using a blend of a polyalkylene oxide triblock copolymer (Pluronic P123, EO₂₀PO₇₀EO₂₀) and *n*-butanol acting as structure-directing agents under mild acidic conditions [15]. The pore structure of KIT-6 consists of two interwoven mesoporous sub-networks quite similar to those of MCM-48 silica. This unique 3D channel network of KIT-6 is thought to provide a highly open porous host with an easy and direct access for guest species, thus facilitating diffusion throughout the pore channels without pore blockage. Moreover, the use of large pore (>5 nm) mesoporous materials exhibiting such an open 3D pore network structure is expected to provide substantially improved adsorption and transport properties as compared to other microporous or mesoporous supports. KIT-6 can adsorb water and other polar liquids because of the expected presence of hydroxyl groups on the pore surface [18]. Due to all these reasons KIT-6 was chosen for the studies reported in this work as a model porous material with the well-defined 3D system of sufficiently large mesopores capable of adsorbing ILs. A sufficient sensitivity for the measurements of carbon dioxide diffusion in the confined CO₂/IL mixture was achieved by applying a high magnetic field of 17.6 T in our PFG NMR studies. In addition to the confined CO₂/IL mixture, also the corresponding bulk IL, bulk CO₂/IL mixture and confined pure IL were studied by PFG NMR.

2. Experimental

2.1. Materials

The sample of KIT-6 silica was synthesized as described in Refs. [16,17]. Briefly, 7.9 g of P123 was dissolved in 287 g of distilled water and 14.6 g of HCl (37.5%) under vigorous stirring at 35 °C. After complete dissolution, 7.9 g of *n*-butanol (99.9%, Fluka) was added to the mixture and after one hour, 20.4 g of TEOS was added at once to the homogeneous solution. The resulting mixture was left under stirring at 35 °C for 24 h. Thereafter the mixture was put in an oven for hydrothermal treatment at 100 °C for another 24 h under static conditions. Silica products were then filtered hot and dried at 100 °C for 24 h. For template removal, the as-synthesized silica powders were first shortly slurred in an ethanol–HCl mixture and were subsequently calcined at 550 °C for 5 h under air flow.

The ionic liquid 1-*n*-butyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide [bmim]⁺[Tf₂N][−] was purchased from Ionic Liquid Technologies (99% purity). To increase the signal-to-noise ratio of ¹³C PFG NMR measurements, ¹³C-enriched CO₂ was used with 99% isotopic purity (Sigma–Aldrich).

2.2. Structural characterization

Nitrogen adsorption isotherms were measured at liquid nitrogen temperature (77.4 K) using an Autosorb-1A automatic volumetric adsorption analyzer from Quantachrome (Boynton Beach, FL, USA). Prior to the measurements, the KIT-6 sample was degassed at 473 K for 12 h under high vacuum. The Brunauer–Emmett–Teller (BET) equation was used to calculate the apparent surface area from adsorption data obtained at P/P₀ between 0.05 and 0.3 and was found to be equal to 814 m²/g. Total pore volume of micro-mesopores was equal to 1.12 cm³/g and was estimated from the amount of nitrogen adsorbed at P/P₀ = 0.95 using the

Gurvitch rule and assuming that adsorption on the external surface was negligible compared to the adsorption in pores. Pore size was determined by applying non-local density functional theory (NLDFT) methods to the desorption branch of the isotherm and considering adsorption of nitrogen at 77 K in cylindrical silica pores [19]. The implemented NLDFT models were supplied by the Quantachrome Autosorb iQ 3.01 software. A main pore size of 8.5 nm was obtained. The measured powder XRD pattern confirmed the expected structure of KIT-6.

2.3. Preparation of NMR samples

The particles of KIT-6 silica were loaded with [bmim]⁺[Tf₂N][−] using the procedure similar to that described in Refs. [8,11]. To facilitate penetration of the IL into mesopores of KIT-6 silica the IL was dissolved in ethanol, added to the silica, and stirred for 1 h at 298 K. The solutions with 75 and 229 mg/ml of the IL in ethanol were used to prepare two confined IL samples. In both cases sufficiently large volume of the mixture was added to the silica to insure that there is enough IL to occupy all intraparticle pores and also all gaps between KIT-6 particles in the resulting ethanol-free samples. Ethanol was evaporated initially in air at 353 K. Following this initial evaporation, the rest of the ethanol was removed by exposing the sample to a high vacuum at 373 K for around 24 h. Essentially the same ethanol removal procedure was used in Refs. [8,11] which report preparation of KIT-6 particles loaded with ethanol-free and water-free [bmim]⁺[Tf₂N][−]. Analysis of the NMR data, which are shown below, confirm that this sample preparation procedure can be used to achieve a full loading of the KIT-6 mesopores with the water-free and ethanol-free IL. In particular, no water and ethanol lines were observed in the recorded ¹H and ¹³C NMR spectra. While the observed line broadening of the IL lines in the samples containing KIT-6 can mask some small contribution of the water and ethanol lines to the measured NMR spectra, such contributions are expected to be seen in PFG NMR measurements because both water and ethanol diffuse much faster than the ions. The absence of fast-diffusing components in the measured PFG NMR data allowed us to conclude that the concentration of water and ethanol in the studied samples was negligibly small. This conclusion was also confirmed by a good agreement of the measured cation diffusivity in the IL between the KIT-6 particles and the corresponding diffusivity in the water-free and ethanol-free bulk IL.

For preparation of the samples for PFG NMR studies, the bulk IL or silica particle beds filled with the IL were placed into 5 mm NMR tubes and connected to a custom-made vacuum system where the samples were made sorbate-free under high vacuum (<10^{−3} Pa) at an elevated temperature (around 373 K) for at least 24 h. CO₂ was loaded into the degassed samples by freezing a known amount of CO₂ into the sample from a calibrated volume of the vacuum system. The height of the samples in the NMR tubes was in all cases close to 20 mm. In all samples containing carbon dioxide the CO₂ concentration was 0.06 CO₂ molecules per cation–anion pair. This concentration was determined from the measured intensities of the IL and CO₂ ¹³C NMR signals in the studied samples in the same way as discussed in our previous paper [20]. The CO₂ concentration was chosen to be sufficiently small so that the ion diffusion in the bulk mixture samples remains to be essentially unperturbed by the presence of CO₂ [20].

2.4. Diffusion NMR studies

PFG NMR studies were performed using a wide bore 17.6 T Bruker Biospin spectrometer equipped with a diff60 diffusion probe (Bruker Biospin) and a Great60 gradient amplifier (Bruker Biospin). Self-diffusivities of the [bmim]⁺ cation and CO₂ were measured,

respectively, using the 13-interval ^1H PFG NMR pulse sequence and the stimulated echo ^{13}C PFG NMR pulse sequence [21–23]. The 13-interval sequence allows reducing or even eliminating the disturbing influence of magnetic susceptibility effects [23]. Such effects can be quite significant in the case of ^1H PFG NMR spectroscopy which was employed for the studies of the cation diffusion. The stimulated echo sequence offers the possibility of reducing the influence of the transverse (T_2) NMR relaxation on the measured signal by keeping short the sequence time intervals during which this relaxation process reduces the measured signal. For this sequence, the absence of disturbing susceptibility effects is expected because this sequence was used only for ^{13}C nuclei characterized by a relatively low gyromagnetic ratio.

Diffusivities were obtained from monoexponential or biexponential fit of the PFG NMR attenuation curves (i.e., the dependences of the normalized PFG NMR signal intensity (S) on the amplitude of the magnetic field gradient (g)) with the following relation [21–23]

$$\Psi \equiv \frac{A(g)}{A(g=0)} = \sum_{i=1}^n p_i \exp\left(\frac{-1}{6} q^2 \langle r_i^2(t) \rangle\right) = \sum_{i=1}^n p_i \exp(-q^2 t D_i), \quad (1)$$

where q is equal to $\gamma\delta g$ and $2\gamma\delta g$, respectively, for the stimulated echo and 13-interval sequence. γ is the gyromagnetic ratio, δ denotes the duration of the magnetic field gradient pulses, and t is the diffusion time defined in Refs. [21–23]. $\langle r_i^2(t) \rangle$ and p_i denote, respectively, the mean square displacement (MSD) and fraction of sorbate molecules diffusing with the diffusivity D_i . An observation of monoexponential behavior ($n = 1$ in Eq. (1)) corresponds to the situation when all molecules or ions under study diffuse with a single diffusivity. In contrast, biexponential behavior ($n = 2$ in Eq. (1)) indicates that there are two ensembles of molecules or ions diffusing with different diffusivities. The expression in the right-hand part of Eq. (1) was obtained using the Einstein relation:

$$\langle r_i^2(t) \rangle = 6D_i t. \quad (2)$$

While gradients up to 30 T/m were used, the majority of the measurements were performed with the maximum gradient of around 10 T/m. The gradient duration was around 3 ms. In a measurement of a single PFG NMR attenuation curve the number of signals averaged for each gradient value was between 160 and

448. The repetition time was approximately a factor of 3 larger than the corresponding longitudinal (T_1) NMR relaxation time. The values of T_1 were measured by the standard inversion recovery (IR) pulse sequence. Carr-Purcell-Meiboom-Gill (CPMG) pulse sequence was used to measure the transverse (T_2) NMR relaxation times. In the CPMG measurements the value of “tau” was 100 μs and the number of echoes was up to 4000. All NMR measurements reported in this paper were performed at 298 K.

3. Results and discussion

Fig. 1 shows examples of the ^1H NMR spectra obtained by one pulse sequence for [bmim] $^+$ [Tf2N] $^-$ confined in the KIT-6 mesoporous silica as well as for bulk [bmim] $^+$ [Tf2N] $^-$.

All the lines in the spectra originate from the [bmim] $^+$ cation. The [Tf2N] $^-$ anion does not have any hydrogen atoms. Hence, there is no contribution of the anion to the spectra in Fig. 1. It is seen that there is a significant line broadening in the spectrum of the confined IL sample in comparison to the spectrum of the bulk IL. Such broadening can be explained by the reduced cation mobility under the conditions of confinement in mesopores and by the magnetic susceptibility effects. The lines in Fig. 1B are assigned as follows. The line at around 8.5 ppm and the two lines between 7.5 and 7.0 ppm originate from the protons of the imidazolium ring. The lines at around 4.1, 1.7 and 1.2 ppm belong to the protons of the butyl chain of the cation. The two strongest lines at around 3.7 and 0.7 ppm originate, respectively, from the methyl group of the butyl chain and the methyl group attached to the imidazolium ring of the cation. The cation diffusivity was obtained from ^1H PFG NMR measurements using the two strongest lines at around 0.7 and 3.7 in the spectra of the samples without KIT-6, and the corresponding two broad lines centered at 1 and 3.8 ppm in the spectra of the samples containing KIT-6. To obtain PFG NMR attenuation curves, the areas under these lines were measured as a function of the gradient strength. For each sample, the PFG NMR attenuation curves presented in this work for the cation show the results of averaging over the corresponding areas under the two narrow lines or the two broad lines discussed above. It was verified that such average attenuation curves coincide (within the experimental uncertainty) with the corresponding attenuation curves measured for the individual cation lines. An area under the single $^{13}\text{CO}_2$ line at 125 ppm was used for ^{13}C PFG NMR diffusion measurements of CO_2

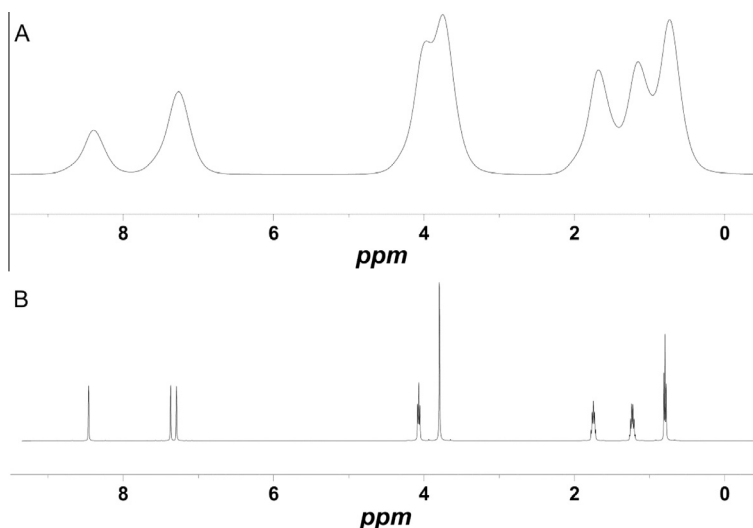


Fig. 1. Examples of the ^1H NMR spectra for the IL/KIT-6 sample (A) and the bulk IL sample (B). The spectra were obtained using a one pulse sequence. The chemical shifts were referenced internally to HDO, viz., D_2O in H_2O mixture.

diffusion. No significant line broadening was observed for this line due to the presence of KIT-6 particles. Signal-to-noise limitations prevented us from performing ^{13}C PFG NMR measurements of ions in a sufficiently large range of the signal attenuation.

In order to check the reproducibility of our diffusion data obtained for the samples containing KIT-6 the measurements were performed with two different PFG NMR samples. The samples were prepared using different amounts of ethanol, but the same KIT-6 silica, and IL or CO_2 /IL mixture (see Section 2 for details). These samples are expected to exhibit the same intraparticle diffusivities because the mesopores in both samples should be fully filled with the identical IL or CO_2 /IL mixture. It was observed that there is an excellent agreement between the PFG NMR attenuation curves measured with these two samples under identical conditions. This is demonstrated in Fig. 2 which presents examples of the attenuation curves measured in the two samples together with the corresponding attenuation curves for the bulk CO_2 /IL mixture.

The results in Fig. 2A and B clearly show that while the attenuation curves measured in the KIT-6 containing samples are very similar, there is a noticeable difference between these attenuation curves and the corresponding results obtained for the bulk sample. To quantify this difference, we obtained the PFG NMR attenuation curves which are averages over the attenuation curves measured under the identical conditions in the two samples which were

prepared using different amounts of ethanol. Fig. 3 shows such average ^1H PFG NMR attenuation curves for the diffusion of the $[\text{bmim}]^+$ cation in the samples with and without CO_2 at different diffusion times.

Fig. 4 presents corresponding average ^{13}C PFG NMR attenuation curves for the diffusion of CO_2 at different diffusion times. In addition to the averaged data for the samples containing KIT-6 silica, the figures also show the corresponding reference attenuation curves measured in the bulk samples.

Fitting the attenuation curves in Figs. 3 and 4 by Eq. (1) reveals that the biexponential fitting function (i.e., Eq. (1) with $n = 2$) is required to fit the data for the confined IL and confined CO_2 /IL mixture samples. This observation points at the existence of two ensembles of the same type of species diffusing with different diffusivities. This is an expected result because there are at least the following two diffusing ensembles in these samples: (i) molecules and/or ions that diffuse inside the mesopores of KIT-6 particles during the diffusion time of the measurement, and (ii) molecules and/or ions that diffuse in the macroscopic volume between the KIT-6 particles and particle aggregates or above the particles bed. In contrast, only one ensemble of the same type of diffusing species is expected in the reference bulk samples. Hence, a monoexponential function was used to fit the attenuation curves for the $[\text{bmim}]^+$

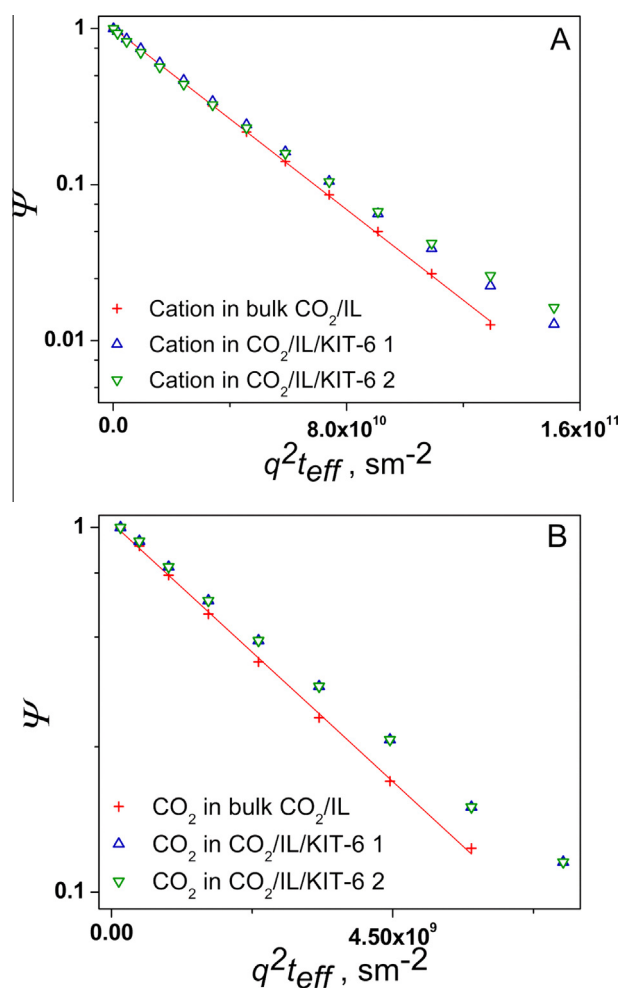


Fig. 2. Examples of PFG NMR attenuation curves measured for the two identical samples of beds of KIT-6 particles saturated with the CO_2 /IL mixture and for the corresponding bulk CO_2 /IL mixture at 298 K. The attenuation curves were obtained for the $[\text{bmim}]^+$ cation diffusion by ^1H PFG NMR at 10 ms diffusion time (A), and for the CO_2 diffusion by ^{13}C PFG NMR at 90 ms diffusion time (B). The lines show the best fit of the data measured in the bulk mixtures using Eq. (1) with $n = 1$.

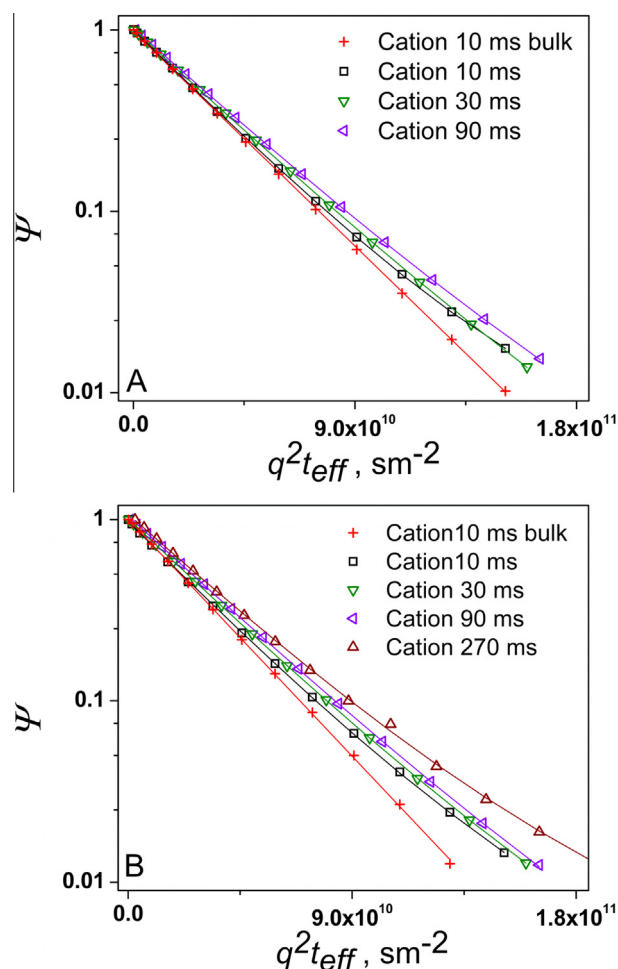


Fig. 3. ^1H PFG NMR attenuation curves for the $[\text{bmim}]^+$ cation diffusion at different diffusion times in a bed of KIT-6 particles saturated with the IL (A), and with the CO_2 /IL mixture (B). The curves were obtained by averaging the corresponding attenuation curves measured at 298 K in the two samples containing identical beds of KIT-6 particles. Also shown for comparison are the ^1H PFG NMR attenuation curves for the $[\text{bmim}]^+$ cation in the corresponding bulk samples. The lines show the results of best fit using Eq. (1) with $n = 1$ for the bulk samples and with $n = 2$ for the samples containing KIT-6 silica.

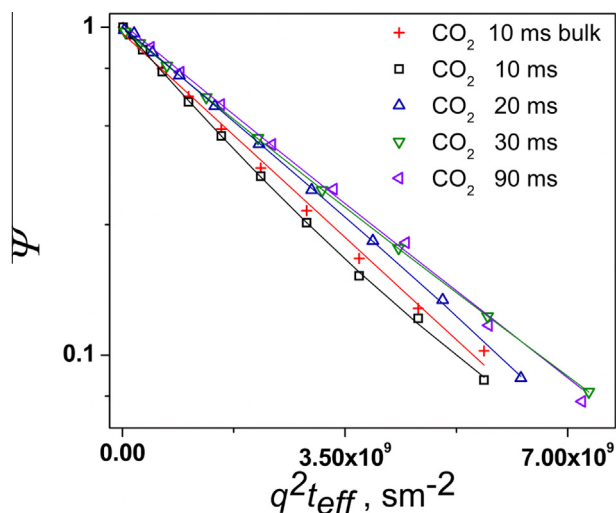


Fig. 4. ^{13}C PFG NMR attenuation curves for the CO_2 diffusion at different diffusion times in a bed of KIT-6 particles saturated with the CO_2/IL mixture. The curves were obtained by averaging the corresponding attenuation curves measured at 298 K in the two samples containing identical beds of KIT-6 particles. Also shown for comparison is the ^{13}C PFG NMR attenuation curve measured in the corresponding bulk CO_2/IL sample. The lines show the results of best fit using Eq. (1) with $n = 1$ for the bulk sample and with $n = 2$ for the samples containing KIT-6 silica.

cation and CO_2 in the reference bulk samples (Figs. 3 and 4). The results of fitting the attenuation curves in Fig. 3A for the diffusion of the $[\text{bmim}]^+$ cation in the confined IL by Eq. (1) with $n = 2$ and in the bulk IL by Eq. (1) with $n = 1$ are shown in Table 1. The corresponding data obtained from fitting the attenuation curves in Fig. 3B for the diffusion of the $[\text{bmim}]^+$ cation in the samples with CO_2 are presented in Table 2. Table 3 shows the results of fitting the PFG NMR attenuation curves for CO_2 diffusion in Fig. 4 by Eq. (1).

The results in Tables 1 and 2 demonstrate that around 90% of all cations in the samples of the confined IL and the confined CO_2/IL mixture diffuse with the diffusivities corresponding to those in the bulk samples. The diffusivity and fraction of this component are denoted, respectively, as D_{inter} and p_{inter} . This component is attributed to the diffusion between the KIT-6 particles and above the particle bed. The remaining component diffuses with the diffusivity which is around a factor of 2 lower than that in the bulk samples. This diffusivity can be attributed to the diffusion coefficient of the $[\text{bmim}]^+$ cations inside the KIT-6 particles, vis. intraparticle diffusivity. To confirm this assignment, the experimentally observed fraction (p_{intra} in Tables 1 and 2) of the component exhibiting the intraparticle diffusion was compared with the expected fraction of the PFG NMR signal corresponding to intraparticle diffusion.

The expected fraction was estimated as follows. The results of T_2 proton NMR relaxation measurements of the confined IL and confined CO_2/IL mixture by the standard CPMG pulse sequence

Table 1

Results of fitting the measured ^1H PFG NMR attenuation curves for the cation diffusion in the bulk IL and IL/KIT-6 samples at 298 K using Eq. (1) with $n = 1$ or 2. The corresponding values of RMSD obtained by Eq. (2) are also shown.

	t , ms	10	30	90
Bulk IL	D , $10^{-11} \text{ m}^2 \text{ s}^{-1}$	3.0 ± 0.4	3.1 ± 0.5	3.1 ± 0.5
IL/KIT-6	D_{inter} , $10^{-11} \text{ m}^2 \text{ s}^{-1}$	3.2 ± 0.5	3.0 ± 0.5	2.9 ± 0.4
	p_{inter}	0.92 ± 0.02	0.91 ± 0.02	0.92 ± 0.02
	$\sqrt{\langle r^2 \rangle_{\text{inter}}}$, μm	1.4 ± 0.2	2.3 ± 0.4	4.0 ± 0.6
	D_{intra} , $10^{-11} \text{ m}^2 \text{ s}^{-1}$	1.7 ± 0.3	1.7 ± 0.3	1.5 ± 0.2
	p_{intra}	0.08 ± 0.02	0.09 ± 0.02	0.08 ± 0.02
	$\sqrt{\langle r^2 \rangle_{\text{intra}}}$, μm	1.0 ± 0.2	1.7 ± 0.2	2.8 ± 0.4

Table 2

Results of fitting the measured ^1H PFG NMR attenuation curves for the cation diffusion in the bulk CO_2/IL and $\text{CO}_2/\text{IL}/\text{KIT-6}$ samples at 298 K using Eq. (1) with $n = 1$ or 2. The corresponding values of RMSD obtained by Eq. (2) are also shown.

	t , ms	10	30	90	270
Bulk CO_2/IL	D , $10^{-11} \text{ m}^2 \text{ s}^{-1}$	3.4 ± 0.5	3.2 ± 0.5	2.8 ± 0.4	
$\text{CO}_2/\text{IL}/\text{KIT-6}$	D_{inter} , $10^{-11} \text{ m}^2 \text{ s}^{-1}$	3.3 ± 0.5	3.1 ± 0.5	3.0 ± 0.5	2.9 ± 0.4
	p_{inter}	0.93 ± 0.02	0.94 ± 0.02	0.93 ± 0.02	0.94 ± 0.02
	$\sqrt{\langle r^2 \rangle_{\text{inter}}}$, μm	1.4 ± 0.2	2.4 ± 0.4	4.0 ± 0.6	6.9 ± 1
	D_{intra} , $10^{-11} \text{ m}^2 \text{ s}^{-1}$	1.4 ± 0.2	1.5 ± 0.2	1.6 ± 0.2	1.1 ± 0.2
	p_{intra}	0.07 ± 0.02	0.06 ± 0.02	0.07 ± 0.02	0.06 ± 0.02
	$\sqrt{\langle r^2 \rangle_{\text{intra}}}$, μm	0.9 ± 0.1	1.6 ± 0.2	2.9 ± 0.4	4.2 ± 0.6

Table 3

Results of fitting the measured ^{13}C PFG NMR attenuation curves for the CO_2 diffusion in the bulk CO_2/IL and $\text{CO}_2/\text{IL}/\text{KIT-6}$ samples at 298 K using Eq. (1) with $n = 1$ or 2. The corresponding values of RMSD obtained by Eq. (2) are also shown.

	t , ms	10	20	30	90
Bulk CO_2/IL	D , $10^{-10} \text{ m}^2 \text{ s}^{-1}$	4.1 ± 0.6	4.1 ± 0.6	3.8 ± 0.6	3.6 ± 0.5
$\text{CO}_2/\text{IL}/\text{KIT-6}$	D_{inter} , $10^{-10} \text{ m}^2 \text{ s}^{-1}$	4.1 ± 0.6	3.9 ± 0.6	3.9 ± 0.6	3.9 ± 0.6
	p_{inter}	0.86 ± 0.05	0.88 ± 0.03	0.89 ± 0.03	0.88 ± 0.03
	$\sqrt{\langle r^2 \rangle_{\text{inter}}}$, μm	5 ± 1	7 ± 1	8 ± 1	15 ± 2
	D_{intra} , $10^{-10} \text{ m}^2 \text{ s}^{-1}$	11 ± 7.6	2.3 ± 0.4	1.0 ± 0.2	0.8 ± 0.1
	p_{intra}	0.14 ± 0.05	0.12 ± 0.03	0.11 ± 0.03	0.12 ± 0.03
	$\sqrt{\langle r^2 \rangle_{\text{intra}}}$, μm	8 ± 3.5	5 ± 1	4 ± 1	7 ± 1

revealed an additional component with a very short T_2 time of around 8 ms. This component was not observed in the bulk IL and CO₂/IL samples. Hence, it is attributed to the [bmim]⁺ cations confined in mesopores. In addition to this component, two other components, which have at least an order of magnitude larger T_2 times, were observed in the samples containing KIT-6. The existence of more than one component with large T_2 times can be explained by the influence of the internal magnetic field gradients [24–26] as discussed in the [supplementary data section](#). Such gradients can be produced by the magnetic susceptibility differences between the relatively short (20 mm) sample and the gas phase above the sample in the NMR tubes as well as between the IL and KIT-6 particles inside the sample (see [supplementary data](#) for more details). The fraction of the T_2 component attributed to the [bmim]⁺ cations confined in mesopores is around 0.3. This value can be compared with the estimate based on the known fraction of the pore volume in a KIT-6 particle (around 0.4) [16]. When no efforts are made to minimize the volume fraction of gaps between the particles of a bed, the packing density of around 0.5 can be obtained for three-dimensional crystal sphere packings that are strictly jammed [27]. Using this packing density as an estimate of the packing density in the KIT-6 beds we obtain $(0.4 \times 0.5) / (0.4 \times 0.5 + 0.5) = 0.29$ for the fraction of the intrapore volume. This value is in a good agreement with the value of 0.3 obtained from the T_2 NMR measurements. In the PFG NMR measurements, the contribution of this component to the measured signal is expected to be smaller than 0.3 due to T_2 NMR relaxation. Under our measurement conditions the total time interval ($2\tau_1$) of the ¹H PFG NMR 13-interval sequence during which the signal is reduced by the T_2 relaxation was around 8 ms. Hence, p_{intra} can be estimated as $0.3 \times \exp(-2\tau_1/T_2) = 0.3 \times \exp(-1) = 0.11$. This value is similar to the experimentally observed values of p_{intra} which confirms our assignment of p_{intra} and D_{intra} to the fraction and diffusivity of the cations diffusing inside the pore network of KIT-6 particles. The T_1 NMR relaxation times were observed to be around 1 s, i.e., much larger than the diffusion times used in the measurements. As a result, the T_1 NMR relaxation process is not expected to influence the fractions of the PFG NMR signal.

The experimentally observed reduction of the cation diffusivity due to the confinement in mesopores can be rationalized by estimating the intraparticle cation diffusivity (D_{intra}) under the conditions of a fast exchange between the surface bound ions and the ions surrounded by other ions in the KIT-6 pores [11]. Assuming that the cation/anion pair can be approximated as a sphere with the diameter of 1 nm [11] we estimate that the volume fraction of the surface-bound ions (p_{surf}) in cylindrical mesopores with the diameter of 8.5 nm is around 0.42. The molecular dynamics (MD) simulations reported in Ref. [8] for the [bmim]⁺[Tf₂N]⁻ confined in the KIT-6 mesoporous silica revealed that the diffusivity of the surface bound cations (D_{surf}) is reduced by more than one order of magnitude in comparison to that of the bulk IL (D_{bulk}). Hence, the contribution of this diffusivity to the mean intrapore diffusion coefficient can be neglected. As a result, D_{intra} can be estimated as $p_{surf} \times D_{surf} + (1 - p_{surf}) \times D_{bulk} \approx (1 - p_{surf}) \times D_{bulk} = 0.58D_{bulk}$. Clearly, this is in a good agreement with our experimental observations (Tables 1 and 2).

The PFG NMR data in Table 3 for the CO₂ diffusion in the confined CO₂/IL mixture can be interpreted in a similar way as that for the [bmim]⁺ cation diffusion. It is seen that the largest component of the CO₂ molecules described by D_{inter} and p_{inter} diffuse with the diffusivity similar to that in the bulk CO₂/IL mixture at all diffusion times used. Hence, this component can be attributed to the CO₂ diffusion in the CO₂/IL mixture located between the KIT-6 particles or above the particle bed. The second CO₂ component represented by D_{intra} and p_{intra} can be attributed to diffusion inside the KIT-6 particles, i.e., intraparticle component. In order to confirm

this assignment we have estimated the expected value of p_{intra} in the same way as it was done for the cations. In complete analogy with the cation data, the results of T_2 carbon-13 NMR relaxation measurements of the CO₂ signal in the confined CO₂/IL mixture by the standard CPMG pulse sequence revealed an additional component with a short T_2 time of around 9 ms. This component was not present in the bulk CO₂/IL mixture. The fraction of this component in the CPMG measurements was around 0.3 and the total time interval ($2\tau_1$) of the ¹³C PFG NMR stimulated echo sequence during which the signal is reduced by the T_2 relaxation was around 8 ms. Hence, p_{intra} can be estimated as $0.3 \times \exp(-2\tau_1/T_2) = 0.3 \times \exp(-0.89) = 0.12$. This estimate appears to be in a good agreement with the values of p_{intra} in Table 3.

It is seen in Table 3 and Fig. 4 that the diffusivity of the smaller (i.e., intraparticle) component of CO₂ molecules decreases with increasing diffusion time. Such diffusivity decrease can be attributed to the restrictions of the diffusion process by the external surface of KIT-6 particles. Similar observations of suppressed intraparticle diffusion due to transport barriers at the external particle surface were reported for a number of porous systems [22]. Fig. S5 in Ref. [16] shows a typical SEM image of the KIT-6 sample. This image suggests that smaller (i.e., primary) KIT-6 particles with sizes in the range between 3 and 8 microns form much larger particle aggregates. The existence of transport barriers in the form of partially blocked pore entrances at the external particle surface can lead to the observed decrease in the value of D_{intra} with increasing diffusion time. Table 3 shows the values of the square root of MSDs (RMSDs) which were determined for CO₂ diffusion using Eq. (2). It is seen that the decrease of D_{intra} of CO₂ molecules is observed for RMSDs comparable with the sizes of primary KIT-6 particles. This observation supports our explanation of the measured dependence of D_{intra} on diffusion time for CO₂. An absence of a comparable decrease of the values of D_{intra} of the [bmim]⁺ cations with increasing diffusion time in the range between 10 and 90 ms (Tables 1 and 2) can be explained by the observation that at these diffusion times the RMSD values for the cation were smaller than both the corresponding RMSD values of CO₂ and the expected average size of the primary KIT-6 particles. The range of the diffusion times and the corresponding values of RMSD for the cation diffusion were limited by the proton T_1 NMR relaxation time (around 1 s). In order to reach RMSDs of the cations similar to those observed for CO₂ at small diffusion times, additional PFG NMR measurements of the cation diffusion were performed for a larger diffusion time of 270 ms, which was comparable with the T_1 relaxation time (Fig. 3 and Table 2). The data in Table 2 is consistent with the existence of a decrease of the intraparticle diffusivity of the cation when the diffusion time is increased up to 270 ms. Hence, as expected, the diffusivity of the cation shows similar trend as that observed for CO₂ when the value of RMSD increases to the extent that it becomes comparable with the sizes of the primary KIT-6 particles.

Proton PFG NMR studies of the [bmim]⁺ cation in pure [bmim]⁺[Tf₂N]⁻ confined in KIT-6 mesopores, which were somewhat larger than those in the sample used in this work, were previously reported in Refs. [8,11]. However, in this previous work the main focus was given to the diffusivities measured by PFG NMR in the limit of large diffusion times. The results reported in Ref. [11] for the fully filled mesopores show an existence of the dependence of the effective cation diffusivity on diffusion time. This dependence was found to be consistent with the observation of the decrease in the value of D_{intra} made in the present work when the diffusion time was increased up to 270 ms (Table 2).

Clearly, the best estimate of the intraparticle diffusivity of CO₂ which is not influenced by the diffusion restrictions at the external particle surface is provided by the value of D_{intra} in Table 3 at the smallest diffusion time of 10 ms. It is seen that, within the large

experimental uncertainty reported in the table, the value of the intraparticle diffusivity of CO₂ can be similar to or even larger than the corresponding CO₂ diffusivity in the bulk CO₂/IL mixture. The results of recently reported MD simulations [9] provide a possible explanation of this observation. The simulations indicate that the IL density can be smaller at the pore walls of silica slit pores in comparison to the corresponding bulk IL density [9]. These simulations also show that the CO₂ diffusivity in the confined IL can be larger than that in the bulk IL. This result was attributed to fast CO₂ diffusion near silica pore walls where the frequency of collisions of CO₂ molecules with ions is reduced in comparison to that in the bulk IL [9]. In the studied KIT-6 particles the IL density at the pore surface can also be reduced. In particular, such density decrease can be caused by the roughness of the pore surface and the relatively large size of the ions. The reduced ion density at the pore walls provides for an additional free volume that can lead to fast CO₂ diffusion near these walls.

4. Conclusions

This work reports microscopic experimental studies of diffusion in a mixture of CO₂ and IL confined in nanopores. The diffusion measurements were performed by high field PFG NMR using high magnetic field gradients up to 30 T/m. The studies were carried out with the mixture of CO₂ and [bmim]⁺[Tf₂N][−] as well as with the pure [bmim]⁺[Tf₂N][−] confined in mesopores of KIT-6 silica. The results of these studies show that (within the relatively large experimental uncertainty of the CO₂ diffusion measurements) the CO₂ diffusivity under the conditions of confinement is similar to or even larger than the corresponding diffusivity in the bulk CO₂/IL mixture. The possible explanation of this observation was discussed. The smaller experimental uncertainty of the cation diffusion measurements allowed observing a decrease of the cation diffusivity under the conditions of confinement in comparison to that in the bulk IL. These data were obtained for the smallest diffusion time of 10 ms used in the measurements. The diffusivities of the cations and carbon dioxide were found to decrease with increasing diffusion time. This behavior is consistent with the diffusion slowdown due to transport resistances at the external surface of KIT-6 particles.

Acknowledgments

We are grateful for the financial support of this work by the NSF – United States award (CBET No. 0967703). NMR data were obtained at the Advanced Magnetic Resonance Imaging and Spectroscopy (AMRIS) facility in the McKnight Brain Institute of the

University of Florida. The NMR measurement time is provided in the framework of the external user program of the National High Magnetic Field laboratory, AMRIS is gratefully acknowledged. Members of the Vasenkov Group would especially like to thank Dr. Dan Plant at AMRIS for his help with NMR measurements.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.micromeso.2014.12.005>.

References

- [1] L.M. Robeson, *J. Membr. Sci.* 62 (1991) 165–185.
- [2] P. Scovazzo, J. Kieft, D.A. Finan, C. Koval, D. DuBois, R. Noble, *J. Membr. Sci.* 238 (2004) 57–63.
- [3] P. Scovazzo, D. Havar, M. McShea, S. Mixon, D. Morgan, *J. Membr. Sci.* 327 (2009) 41–48.
- [4] R.E. Baltus, R.M. Counce, B.H. Culbertson, H.M. Luo, D.W. DePaoli, S. Dai, D.C. Sep, *Sci. Technol.* 40 (2005) 525–541.
- [5] J.J. Close, K. Farmer, S.S. Moganty, R.E. Baltus, *J. Membr. Sci.* 390–391 (2012) 201–210.
- [6] A.I. Labropoulos, G.E. Romanos, E. Kouvelos, P. Falaras, V. Likodimos, M. Francisco, M.C. Kroon, B. Iliev, G. Adamova, T.J.S. Schubert, *J. Phys. Chem. C* 117 (2013) 10114–10127.
- [7] J. Rodriguez, M.D. Elola, D. Laria, *J. Phys. Chem. C* 116 (2012) 5394–5400.
- [8] S. Li, K.S. Han, G. Feng, E.W. Hagaman, L. Vlcek, P.T. Cummings, *Langmuir* 29 (2013) 9744–9749.
- [9] W. Shi, D.R. Luebke, *Langmuir* 29 (2013) 5563–5572.
- [10] W. Shi, D.C. Soreescu, *J. Phys. Chem. B* 114 (2010) 15029–15041.
- [11] K.S. Han, X. Wang, S. Dai, E.W. Hagaman, *J. Phys. Chem. C* 117 (2013) 15754–15762.
- [12] C. Iacob, J.R. Sangoro, W.K. Kipnusu, R. Valiullin, J. Karger, F. Kremer, *Soft Matter* 8 (2012) 289–293.
- [13] S.M. Chathoth, E. Mamontov, S. Dai, X. Wang, P.F. Fulvio, D.J. Wesolowski, *EPL* 97 (2012).
- [14] L.A. Banu, D. Wang, R.E. Baltus, *Energy Fuels* 27 (2013) 4161–4166.
- [15] F. Kleitz, S.H. Choi, R. Ryoo, *Chem. Commun.* (2003) 2136–2137.
- [16] T.W. Kim, F. Kleitz, B. Paul, R. Ryoo, *J. Am. Chem. Soc.* 127 (2005) 7601–7610.
- [17] F. Kleitz, F. Berube, R. Guillet-Nicolas, C.-M. Yang, M. Thommes, *J. Phys. Chem. C* 114 (2010) 9344–9355.
- [18] Z. Zhang, H. Wang, X. Chen, R. Xie, P. Gao, W. Wei, Y. Sun, *Adsorption* 20 (2014) 883–888.
- [19] J. Landers, G.Y. Gor, A.V. Neimark, *Colloids Surf. A* 437 (2013) 3–32.
- [20] E.D. Hazelbaker, S. Budhathoki, A. Katiyar, J.K. Shah, E.J. Maginn, S. Vasenkov, *J. Phys. Chem. B* 116 (2012) 9141–9151.
- [21] P.T. Callaghan, *Principles of NMR Microscopy*, Clarendon Press, Oxford, 1991.
- [22] J. Kärger, D.M. Ruthven, D.N. Theodorou, *Diffusion in Nanoporous Materials*, Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim, Germany, 2012.
- [23] R.M. Cotts, M.J.R. Hoch, T. Sun, J.T. Markert, *J. Magn. Reson.* 83 (1989) 252–266.
- [24] B.P. Hills, K.M. Wright, J.E.M. Snaar, *Magn. Reson. Imaging* 14 (1996) 715–718.
- [25] M.M. Britton, R.G. Graham, K.J. Packer, *Magn. Reson. Imaging* 19 (2001) 325–331.
- [26] M.M. Britton, R.G. Graham, K.J. Packer, *J. Magn. Reson.* 169 (2004) 203–214.
- [27] S. Torquato, F.H. Stillinger, *J. Appl. Phys.* 103 (2008).